

# Legend

Formalized  
Result

Formalized  
Lemma

Conditional  
Bridge

Definition

Open /  
Pending

## Model Definitions

Utilities, policies, fairness

### Lemma 1 (App. C)

$J_{\min}^* > 0$

### Lemma 2 (App. C)

Item-fairness LP epigraph

### Proposition 2

Symmetric optimum exists

### Reduction Bridges

Original  $\leftrightarrow$  reduced model

### Theorem 3

First-half price monotonicity

### Problem 6

Opposing-type  
LP translation

### Lemma 16 (App. E)

$q_j$  / midpoint algebra

### Problem 11 (App. E)

Three-type LP,  $\gamma = 1$   
bridge, real-BFS interface

### Problem 11

**Closed Witness**  
Dual bound, tight  
primal, pivot existence,  
basic support

### Lemma 4 (App. D)

Basic support +  
no-gap structure

### Lemma 5 (App. D)

Closed-form policy family

### Lemma 12 (App. E)

Symmetric reduction into  $S'$

### Lemma 17 (App. E)

No right-half type-zero mass

### Lemma 13 Perturbations

$x$ -gap and  $z$ -gap certificates

### Lemma 13 (App. E)

Pivot support from  
equalized optimum

### Lemma 15 (App. E)

Coordinates from  
closed/equalized optimum

### Lemma 14 (App. E)

Equality-form optimum  
uniqueness

### Theorem 4 No-Extremes

Equality-form Problem 11  
bridge

### Cold-Start Bounds

No-extreme  $\Rightarrow$  low utility

### No-Fairness Side

Estimated  $\gamma = 0$  benchmark

### Theorem 4 Fairness Wrappers

Source cold-user policy;  
estimated lift; needs  
Problem 11 LP-selection

### Value-Vector Witness

Geometric  $v_j = r^j$   
construction

### Theorem 4 (App. E)

Open: LP-selection/unique  
optimum + source  
no-fairness wrapper

### True-Model Lower Bound

Original two-type half model  
gives  $U_{\min}^*(1, w) > 1/n$